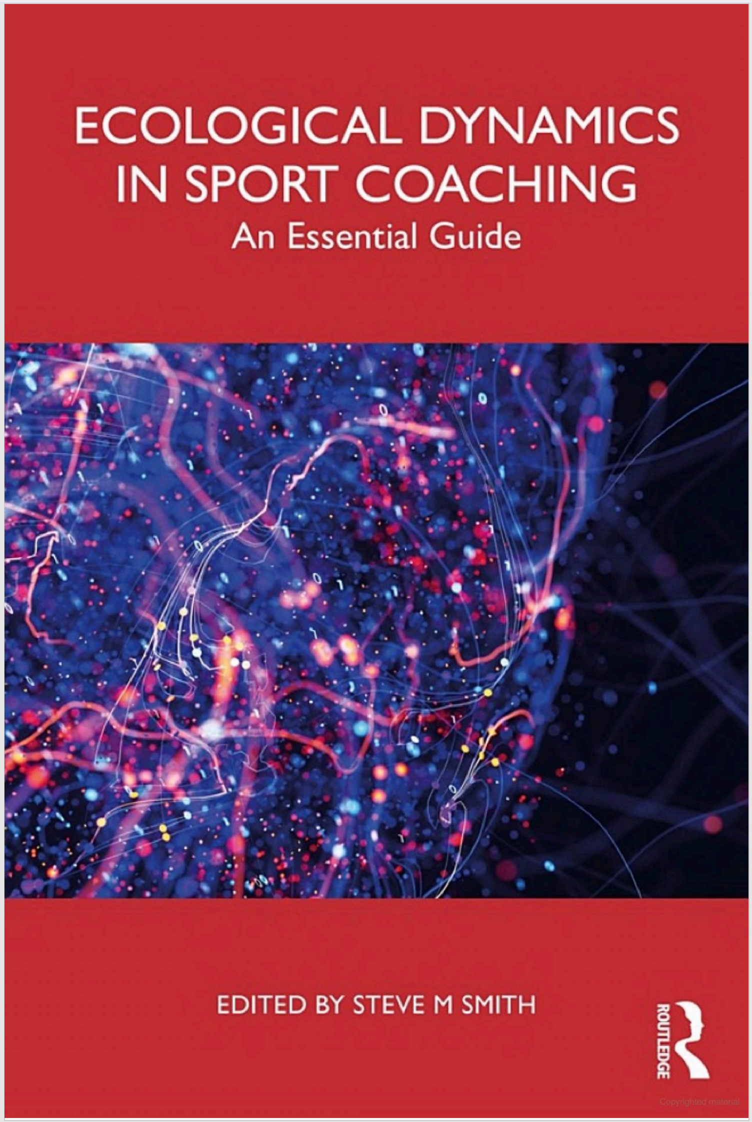
ECOLOGICAL DYNAMICS
IN SPORT COACHING
An Essential Guide
EDITED BY STEVE M SMITH
ROUTLEDGE

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# CONTENTS

**PART 3**
**Coaching Individual Sports**

# TABLES

# PREFACE

Sport is chaotic. Whether it is the unpredictable bounce of a cricket ball off the wicket or the constantly shifting positions and decisions of players in a football

We acknowledge that traditional coaching methods are not inherently wrong. Many have led to successful outcomes. But research increasingly suggests that certain aspects of learning, particularly the transfer of skill from practice to competition, are more effectively supported by principles of ecological dynamics. This book offers a flexible framework to underpin coaching decisions. Coaches can adopt its principles fully or begin by exploring individual elements. It provides a philosophy, a way of thinking about skill and learning that encourages coaches to view the sport itself as the teacher and the environment as central to skill formation.

The structure of the book reflects its pedagogical stance, guiding readers from foundational concepts to applied examples across a variety of sporting contexts. It is organised into three main parts. The first part introduces the core concepts, frameworks, and philosophical underpinnings of ecological dynamics. This opening provides essential grounding for readers, ensuring they are equipped to interpret and implement the ideas presented throughout the book. Following the foundational part, the book explores the application of ecological dynamics across both team and individual sports. Chapters are written by experienced coaching practitioners and scholars, examining how ecological principles manifest within the collective demands of competition, highlighting the complexity of player interactions, tactical structures, and the importance of designing representative learning environments. These chapters delve into how individuals adapt to moment-by-moment variability, and how coaches can support the development of perceptual attunement and action regulation in contexts where skill expression is often highly personalised and situational.

Each sport-specific chapter follows a consistent format. First, the dynamics of the sport are explored, highlighting the unique constraints, patterns, and performance demands that athletes face in competition. This serves as a foundation for designing representative practice tasks. For example, understanding typical work-to-rest ratios or perceptual demands allows coaches to design sessions that mirror the realities of play. The chapters then identify general coaching considerations within an ecological dynamics framework before moving into applied examples and design strategies. Notably, this book does not pretend to offer fixed lesson plans. The nature of ecological dynamics requires coaches to be creative, to shape practice environments based on their unique context, players, goals, and sport demands. As such, each chapter concludes with reflective questions and key takeaways to support ongoing development and contextual application.

While the theoretical principles of ecological dynamics are consistent across sports, their application in practice is deeply sport-specific. However, the diversity of contexts presented in this book offers valuable opportunities for cross-sport learning. Coaches may find that ideas rooted in one sport spark fresh insights for another. For example, the chapter on coaching cricket cautions against over-prioritising representativeness at the expense of learning and challenge, an important consideration that extends far beyond cricket. The chapter on coaching rugby explores how sociocultural influences, including national identity, shape team learning dynamics, raising broader questions about how context influences coaching across cultures. In the chapter on coaching golf, the author highlights the critical role of anxiety in shaping performance, underscoring the importance of embedding psychological realism into practice, something just as relevant in high-pressure moments of any sport. Across these chapters, readers are encouraged not only to engage deeply with their own sport but also to remain open to the transferable lessons offered by others.

In every case, we emphasise that sport should not be reduced to isolated skills or idealised technical models. For many coaches, the instinct to isolate skills stems from a desire to gain control over the messiness of performance, to simplify the coaching process, create tidy progressions, and minimise unpredictability. But while isolating skills may feel more manageable, it often comes at a cost. Skills are not fixed units to be mastered in sterile settings and then transferred into the game; they are emergent solutions, shaped by the dynamic relationship between the athlete and their environment. Even the moments before and after a skill execution, including how players position themselves, perceive information, and adapt in real time, are integral to its function. When skills are removed from this broader system in an attempt to simplify learning, we risk developing athletes who appear technically sound in training but struggle to perform under the unpredictable demands of competition.

A traditional model, where athletes develop technique in artificially structured settings and later attempt to adapt it to game situations, often works only for those with high adaptability; those who are able to stitch isolated skills back into the competitive environment of the sport. This approach may inadvertently exclude athletes who struggle to make that leap from isolated practice to integrated

answers, but it offers better questions. Questions that start with the chaos of sport, not as a problem to be solved, but as the very medium through which skill emerges.

# ACKNOWLEDGEMENTS

This book would not have been possible without the generous contributions of time, insight, and creativity from our sport-specific chapter authors. Their commitment to advancing knowledge in ecological dynamics and sport coaching is at the heart of this book.

I am grateful to the athletes, coaches, and practitioners whose work continues to inspire deeper inquiry into how learning, adaptation, and creativity emerge in sport.

A special thank you to Eden Smithbury, whose expertise in digital design helped bring complex ideas to life. Her creative interpretation of visual concepts added clarity and impact to the book's illustrative content.

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# PART 1
# Foundations of Ecological Dynamics in Sport Coaching

## 1
## ECOLOGICAL DYNAMICS IN SPORT COACHING
### Principles and Practice Foundations

*Steve M Smith*

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The role of a sport coach is to support and develop individuals and groups across environments that range from recreational participation to elite professional performance. Sport coaches work with individuals of varying ages, genders, and physical abilities, within diverse settings and cultural contexts, with no two experiences ever being the same. Despite these differences, a fundamental goal of all sport coaches is to facilitate the acquisition and development of skills to enhance sporting performance. Within the skill acquisition process, the coach must consider the individuality of the learner, the nature of the task to be performed, and the environmental conditions in which the learner aims to execute the skill (Lawrence et al., 2013). The coach is then responsible for designing and delivering a learning environment that effectively supports the development of sporting skills (Martens & Vealey, 2024). This chapter defines ecological dynamics in the context of sport coaching, offering coaches an understanding of the framework and its influence on all aspects of learning, practice, and development.

### Approaches to Skill Acquisition and Development

The dominant skill acquisition theories that have underpinned sport coaching practice in recent decades often conceptualise humans as functioning similarly to computers, where learned skills are mechanically encoded and deployed at appropriate moments. According to these models, skill acquisition occurs through distinct, linear stages, namely, the cognitive, associative, and autonomous stages (Fitts & Posner, 1967), and is supported by many hours of repetitive, sequential practice (DeKeyser, 2020). Within this framework, the sport coach's role is primarily to create an environment in which specific movement patterns required by the sport can be developed. For instance, when coaching kicking in football (soccer), technical proficiency can be judged by observing the movement of joints at the hip, knee, and ankle, and the quality of ball contact (Lees et al., 2010).

Information Processing Theory (IPT), a popular skill acquisition model in sport coaching, conceptualises human learning through a computerised lens (Cassidy et al., 2023). In this model, environmental information is first detected, such as visually tracking a tennis ball after an opponent's shot, then processed in the central nervous system (CNS), where short- and long-term memories are searched to find an appropriate motor representation (e.g., a forehand topspin shot from the centre of the court). The relevant motor programme is then selected and executed as a mechanical action. IPT has provided valuable explanatory power in understanding how perception, memory, and action may be connected (Schmidt, 1975).

These motor programmes, or schemata, consist of invariant features that allow skilled performers to respond more automatically to sensory input (Hodges & Williams, 2019). Schmidt's schema theory refined this view by suggesting that generalised motor programmes (GMPs) are not rigid templates but adaptable structures, modified through parameters such as force, timing, and amplitude. Crucially, schema theory proposed recall and recognition schemata that enable athletes to generate novel movements without needing a separate program for every possible action (Schmidt, 1975). This was a direct attempt to solve the "storage" and "novelty" problems that earlier closed-loop theories could not explain.

Nevertheless, as these invariant features are considered constant, GMPs (such as a tennis forehand topspin) are typically taught and practised in isolation from environmental variations, adhering to strict technical models dictated by the coach (Haibach-Beach et al., 2023). For example, a forehand shot might be broken down

This reductionist approach frames sport coaching as a linear process, with athletes progressing through clearly defined stages that can be easily controlled, monitored, and assessed over time. Although IPT and schema theory contributed substantially to understanding memory-based control of action, their widespread interpretation in coaching has too often promoted an internalist and centralised view of skill acquisition. The emphasis on internalised programs places the CNS as the dominant decision-maker, inadvertently downplaying the adaptive, emergent, and environment-dependent nature of skilled movement.

Despite its influence, IPT's emphasis on the CNS as the central hub of decision-making and motor control overlooks the crucial role of the peripheral nervous system (PNS) in movement execution. While IPT places memory and cognition at the centre of motor learning, Teques et al. (2017) advocate for a brain–body–environment system in which metastable dynamics (transient stabilisation of different movement patterns) enable athletes to continuously interact with affordances (opportunities for action) in their environment. For example, in a 1v1 basketball scenario, a player may hover between shooting, passing, or driving, with their body poised to adapt depending on the defender's actions. Elite performers can make such rapid decisions due to enhanced sensory capacities within the PNS (Zemková, 2022), indicating that the CNS is not solely responsible for processing information and initiating movement. Instead, athletes often develop intuitive behaviours in response to environmental cues, similar to the reflex of withdrawing one's hand from a hot surface, where the PNS initiates the response without central processing.

Further support for the role of the PNS in motor control comes from Oishi and Maeshima (2004), who found that autonomic responses such as heart rate and respiration, which are key components of the PNS, differed between elite and non-elite athletes during motor imagery tasks. This does not discount the importance of the CNS in sport but rather underscores that it operates within a larger system that includes the PNS. When learning new skills, the CNS adapts muscle activation patterns to optimise use of the musculoskeletal system (Shemmell et al., 2005), highlighting the essential involvement of the PNS in the learning process. Thus, skill acquisition is not purely a cognitive process but a dynamic interaction between sensory inputs, motor responses, and the performance environment. These rapid, context-specific behavioural responses suggest that linear, centrally focused models such as IPT may fall short in accurately explaining how skills are acquired and executed in sport.

IPT-based approaches favour reductionist and linear methods of coaching that rely on technical models grounded in fixed movement patterns. However, no two sporting situations are ever truly identical. In football, for instance, shooting is a key skill that is often broken down into its component parts and practised in isolation. Technical guidance may include instructing the player to approach the ball at an angle (~43°) to optimise balance and force generation, plant the support leg firmly, swing the kicking leg forward with hip and knee flexion and ankle plantarflexion to build speed, then rapidly extend the knee and maintain a firm ankle just before contact to strike the ball with the dorsum of the foot (Lees et al., 2010). While coaches may attempt to increase representativeness by changing shooting locations, adding a goalkeeper, or introducing varied scenarios, this reduction of a complex skill within a dynamic environment neglects essential components of the performance context. It removes the perceptual and decision-making demands required to first create shooting opportunities in contested spaces, evade defenders, and accurately target areas of the goal beyond the goalkeeper's reach (Smith et al., 2025).

By eliminating variability from the learning process, athletes are deprived of essential opportunities to engage with the complexity of real-world sport. Practising movements in isolation from environmental conditions, such as an opponent's location or the movement of the ball, can hinder decision-making and degrade movement effectiveness in competition (Araújo et al., 2006). While these methods may promote technically accurate and repeatable movements, they lack the adaptability required in dynamic sporting contexts (Smith et al., 2023). In the tennis example mentioned earlier, successful shot selection depends on multiple interacting factors: the opponent's hitting power, their court positioning, the trajectory and speed of the ball, and the timing of its bounce. A forehand topspin shot practised in isolation, devoid of such contextual variability, is unlikely to be effective in this real-time scenario. Therefore, movement responses must be grounded in the affordances and constraints of the performance environment. As a result, skill acquisition and development models that reflect the complexity, variability, and contextual demands of competitive sport are necessary for more representative and effective coaching practice.

## The Ecological Dynamics Framework

The ecological dynamics framework promotes synergy between behavioural and cognitive approaches to decision-making in sport (Araújo et al., 2006). It is grounded in the principles of ecological psychology and dynamical systems theory. Ecological psychology emphasises the role of perceptual information (e.g., visual cues) in guiding movement, leading to perception–action coupling and affordances (Lobo et al., 2018). Dynamical systems theory, in turn, views athletes as complex, adaptive systems whose movement behaviours emerge from the interaction of multiple constraints over time (Torrents & Balagué, 2006). From this perspective, sport becomes a dynamic battle between opposing systems, where players seek to create temporary asymmetries or perturbations, such as exploiting a defender's misstep or poor positioning, to gain a competitive advantage.

When engaging in sport, individuals undertake actions based on their perceptions just received the ball will have their actions constrained by factors such as the position of teammates (e.g., who is open to pass to), the intentions of defenders (e.g.,

be explicitly recognised in practice design. The coach, as the architect of learning, plays a critical role in manipulating task and environmental conditions to support adaptive skill development. For example, an athlete who is short in stature and possesses limited agility and vertical jump ability (organism constraints) may find it more difficult to access affordances for close-range shooting opportunities in basketball. By designing practice environments that reflect these interacting constraints, coaches enable athletes to develop functional behaviours that are context-sensitive and adaptable to real-world performance demands.

### *Perception–Action Coupling*

Perception–action coupling is a core principle within the ecological dynamics framework, emphasising the reciprocal and continuous relationship between perception and action in shaping athletic behaviour. An athlete perceives an opportunity to act (an affordance), and through their action, new affordances are revealed, creating a cyclical process of interaction with the environment. This ongoing loop asserts that effective sport performance emerges from the dynamic interplay between the athlete and their surroundings, where perception informs action and action, in turn, shapes perception (Mann et al., 2007; Stone et al., 2020). In the context of sport coaching, this concept underscores the value of designing training environments that reflect the perceptual and action demands of competitive performance. Through representative learning design (see Chapter 2), coaches can create practice tasks that simulate the complexity and contextual nuances of competition, thereby supporting more effective skill acquisition (Woods et al., 2019; Headrick et al., 2015).

Research illustrates that athletes who develop a more sophisticated perception of their environment can better identify and exploit affordances, which are crucial for decision-making and skill execution (Seifert et al., 2016). As elite athletes progress in their training, they become more adept at tuning their perceptual systems to detect these affordances. This enhanced perceptual skill enables them to execute actions more effectively and adaptively, thereby improving their performance (Seifert et al., 2016). Coaches, therefore, have a vital role in facilitating environments that promote the development of perceptual skills necessary for effective action, creating situations where athletes can engage with the dynamic constraints of their sport (Woods et al., 2020).

### *Affordances*

Affordances are a central concept within the ecological dynamics framework, representing the actionable possibilities available to an athlete based on the continuous interplay between perceptual information and individual capabilities (Button et al., 2020b). Rather than being purely external cues, affordances encompass both external perceptions, such as a rugby player choosing to secure the ball rather than pass because they are being tackled, and internal perceptions of ability, reflecting the athlete's skill level and confidence in executing certain actions (Woods et al., 2020). For example, higher-level rugby players may perceive a passing opportunity earlier and execute it effectively before contact is made, illustrating how expertise expands the range and complexity of affordances available.

In practical terms, affordances are opportunities for action that the environment offers an individual, relative to their unique capabilities. This means that the environment does not present fixed possibilities; instead, it invites actions that are shaped dynamically by the interaction between the athlete and their surroundings. For instance, a gap between defenders affords a chance to dribble through for one player, while another player with different abilities, positioning, or confidence may not perceive the same opportunity.

The concept of affordances highlights that effective performance emerges not simply from reacting to environmental stimuli but from perceiving and exploiting these possibilities in a way that is attuned to one's own physical and cognitive capacities. Consequently, affordances are inherently relational; they depend on

variability in competitive play, and respond to constraint variations, including physical condition changes, opponent behaviours, and broader social influences. Notably, within such systems, movement patterns emerge organically without the need for explicit instruction.

Stability is frequently observed within sport performance systems. For instance, two tennis players engaged in an extended baseline rally. However, it is often instability, triggered by system perturbations, that leads to successful performance outcomes. A drop shot during a baseline exchange in tennis is one such example: it disrupts the rally's rhythm and ends the point. Perturbations are defined as actions or events that interrupt the typical flow and coordination patterns of an athlete or team by introducing instability (Robins & Hughes, 2020). The perturbations that create instability are more obvious in individual sports but are also present in team sports. Because these disruptions are often central to achieving success in sport, coaching practices that overly rely on stability, such as isolated skill drills repeated under constant and unchanging conditions, fail to support the learning and refinement of essential destabilisation skills. Therefore, adopting a dynamical systems perspective enables coaches to better design training environments in which athletes can explore and practise both stabilisation (typically defensive) and destabilisation (typically offensive) strategies within game-representative contexts.

The key application of a dynamical systems approach in the learning and practice of sport movements lies in the necessity for team members to function collectively in solving problems and developing an understanding of both individual and collective behaviours within the team system. This process cannot be effectively achieved through the isolation of movements or the separation of sport elements. For example, practising a smash shot in badminton without the preceding sequence of shots that created the opportunity to execute it removes the contextual information and decision-making processes that underpin its effective use in a rally.

Consequently, it is essential that regular team units, such as defensive or offensive sub-groups, develop an awareness of each other's tendencies and capabilities. This shared understanding enables players to recognise the affordances available to one another and anticipate likely movement behaviours when faced with unique or unpredictable situations. Such understanding can be effectively supported through game-based practice and by providing opportunities for players who operate in similar areas of the pitch and fulfil comparable roles to explore solutions collaboratively within representative training environments.

## The Impact of Ecological Dynamics on Sport Coaching

The sport coach is responsible for designing, delivering, and evaluating coaching sessions that enable individuals to learn and practise sporting skills (Côté & Gilbert, 2009). As all actions in sport are intrinsically linked to the unique environmental conditions in which they occur, learning and practising skills without relevant perceptual cues can result in maladaptive learning behaviours (Pinder et al., 2011a). Although isolated practice may enhance the internal mechanical execution of movements, these actions remain disconnected from the environmental perceptions that typically trigger them.

Research comparing movement behaviours in isolated sporting actions versus those performed during competitive match play has consistently identified significant discrepancies. For example, in badminton, when the direction and timing of the shuttlecock are made predictable, such as by removing the opponent's hitting action and the inherent uncertainty surrounding shot type and trajectory, the biomechanics and mechanical load involved in the return movement differ markedly from those observed under match conditions (Smith et al., 2022, 2023). Similarly, in cricket, when a bowling machine is used in place of a live bowler, key perceptual cues, such as the run-up, arm swing, and grip, are eliminated, leading to changes in the batter's movement patterns due to disrupted anticipation and timing processes (Pinder et al., 2009). Further, alterations in player numbers and pitch dimensions in invasion games like football have also been shown to influence players' movement and visual behaviours (Smith & Conway, 2025). Collectively, these findings emphasise that movements learned and practised in isolation, or with limited relevant perceptual cues, lack representativeness and fail to reflect the decision-making demands of actual sporting performance.

In these cases, isolated skill practice may not only produce a biomechanically different movement pattern but also create an additional learning burden for the athlete, who must then "stitch" the isolated skill back into the performance context. This process privileges individuals who are particularly good at reintegrating technical movements under pressure. However, this ability to convert isolated actions into context-specific responses is not necessarily the same as genuine game intelligence or perceptual attunement. Therefore, traditional coaching may inadvertently reward athletes for their adaptability in stitching isolated skills into competition contexts, rather than for emergent decision-making or perceptual sensitivity to affordances. For example, an adaptive player may recognise when to

linear approach risks neglecting such critical elements, particularly visual scanning behaviours essential to effective passing (Aksum et al., 2021). Thus, passing is not simply a bodily movement, it is a context-sensitive response to environmental constraints that must be integrated throughout each stage of learning.

Ecological dynamics supports a non-linear pedagogical approach to coaching (Chow et al., 2021). Rather than progressing in a fixed sequence, learners are immersed in the complexity of the skill from the outset. While this approach often presents challenges in measuring and tracking progress, it better reflects the realities of dynamic sport environments. Coaches may resist non-linear pedagogy due to concerns over how it is perceived by others, such as the slower appearance of tangible outcomes or uncertainty about when and how to intervene with questions and feedback (Stone et al., 2021). In linear models, athletes are highly regulated, and the outcomes rarely deviate from predefined objectives; an approach that limits exploration and independent problem-solving (Härkki et al., 2021).

Linear pedagogy tends to promote standardised movement models (e.g., the forward drive shot in cricket is executed the same way regardless of context) and relies on prescriptive instruction, restricting individuals from developing movement solutions aligned with their own perceptual abilities and physical characteristics (Seifert et al., 2018). By contrast, non-linear approaches embrace the reality that skill development involves fluctuations in learning and performance, not steady and predictable progress. Skills are embedded through engagement with the competitive sport environment itself (Correia et al., 2019).

Non-linear pedagogy can be operationalised through a constraints-led approach, which supports the continuous integration of perception, decision-making, and action (Davids et al., 2013). This approach manipulates constraints to drive adaptation within the system, leading to new patterns of interaction between its components (--Renshaw et al., 2019). These components can include the movements of individual athletes or the collective behaviours of a team. Constraints must be carefully selected by coaches to ensure fidelity to the sport context. They are not limited to technical demands (e.g., applying defensive pressure to a netball shooter) but also encompass tactical (e.g., group responses to opposition tactics in football) and psychological elements (e.g., managing anxiety during putting in golf). To embed such constraints within practice, representative learning designs are employed to simulate environments where realistic movement behaviours and perception–action coupling emerge (Pinder et al., 2011b). These designs should prioritise an individual's exploration of movement solutions rather than enforcing predetermined outcomes dictated by the coach (Chow et al., 2023).

## Key Takeaways

- **Skill Acquisition is Non-Linear**. Traditional linear models over-simplify learning. Ecological dynamics emphasises that skill development is dynamic, context-dependent, and fluctuates rather than progresses predictably.

- **Perception–Action Coupling is Critical**. Effective performance depends on athletes perceiving environmental cues (e.g., opponent movements, ball trajectory) and adapting actions accordingly. Coaches must design training that integrates perception and movement.
- **Affordances Drive Decision-Making**. Athletes act based on perceived opportunities (affordances) in their environment, shaped by their own abilities and those of others. Coaching should enhance athletes' ability to detect and exploit these affordances.
- **Constraints Shape Behaviour**. Individual, task, and environmental constraints (e.g., fatigue, rules, opponent pressure) influence movement solutions. Coaches can manipulate constraints to guide adaptive learning.
- **Dynamical Systems Theory**. Teams self-organise in response to constraints (e.g., opponent tactics, space). Coaches should foster collective problem-solving in game-like contexts rather than rigid positional drills.
- **Instability Drives Adaptation**. Competitive success often comes from disrupting stability in opponents. Training should include variability and perturbations to develop adaptability.
- **Non-Linear Pedagogy Overrides Prescriptive Coaching**. Instead of enforcing rigid techniques, coaches should encourage athletes to explore movement solutions tailored to their strengths and context.
- **Whole Body Systems in Skill Execution**. Movement is not just cognitively controlled; the peripheral nervous system (PNS) enables rapid, intuitive responses. Training should develop sensory attunement.
- **Practice Must Be Context-Sensitive**. Skills reduced to component parts and learned in isolation often fail in real games. Coaches should ensure training aligns with the perceptual and tactical demands of competition.

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